

P1276R0

Void Main

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Abstract

The current entryptoint for C++ is bizarre, special cased, and violates our own best practices. Let's make void a valid return type for main.

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§ 1. Revision History

§ 1.1. Revision 0

Initial Release 🎉

§ 2. Motivation

Main is a sweet special little gremlin in C++. It has interesting properties regarding exceptions and noreturn functions, `argv` is actually deprecated on some unicode aware platforms and it even lets us ignore the return type entirely. And yet, this is the function we introduce to beginners from day one. This magic, bizarre, entry point to our programs is a nightmare. This proposal attempts to solve this, while also standardizing existing extensions *and* archaic literature and examples.

This paper currently works in tandem with the [Desert Sessions](#) proposal, which attempts to solve issues regarding `argv` and the lack of modern support for reading from environment variables. However, it is not a prerequisite of adding the [\[p1275\]](#) API into the standard, nor does [\[p1275\]](#) require that this paper be added to the standard.

§ 3. Design

The idea behind `void main()` is to simply state that a `main()` that takes no arguments can choose to return nothing as well. Given that we currently permit this *with* the `int` returning main signatures this will not break existing code.

§ 4. Wording

All wording is relative to [\[N4762\]](#)

In Section 6.8.3.1 `main` function [**basic.start.main**] paragraph 2, insert

²An implementation shall not predefine the `main` function. This function shall not be overloaded. Its type shall have C++ language linkage and it have a declared return type of `either` type `int` `or` type `void`, but otherwise its type is implementation-defined. An implementation shall allow:

1. ^(2.1) — a function of `()` returning `void` and
2. ^(2.1) ^(2.2) — a function of `()` returning `int` and
3. ^(2.2) ^(2.3) — a function of `(int, pointer to pointer to char)` returning `int`

In paragraph 5, insert:

⁵A `return` statement in `main` has the effect of leaving the main function (destroying any objects with automatic storage duration) and calling `std::exit` with the return value `or 0 in the case of no return value` as the argument. If control flows off the end of the *compound-statement* of `main`, the effect is equivalent to a `return` with operand `0` (see also 13.3), `or calling std::exit(0) once main's scope has exited`

§ References

§ Informative References

[N4762]

Richard Smith. [Working Draft, Standard for Programming Language C++](https://wg21.link/n4762). 7 July 2018. URL: <https://wg21.link/n4762>

[P1275]

Isabella Muerte. [Desert Sessions: Improving Hostile Environment Interactions](https://wg21.link/p1275r0). URL: <https://wg21.link/p1275r0>